Towards AI-Enabled Nuclear Astrophysics

A Research Proposal Report

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Introduction: Why AI?

Nuclear astrophysics is suitable for AI due to some characteristics.

- **Interdisciplinary with Knowledge Gaps**

  Relevant knowledge is distributed across particle and nuclear physics, astrophysics, and more fundamental theories.

- **Simulation-Based & Computationally Constrained**

  Nucleosynthesis networks involve $\sim 10^3$ isotopes, up to $\sim 10^4$ reactions coupled to 2D/3D hydrodynamics of astrophysical environments.

- **Experimentally Expensive**

  Experiments involve many coupled setup parameters, while beam time and machine access are limited and costly.

- **Uncertainty-Sensitivity Intensive**

  Make predictions and identify which uncertain inputs dominate the results should be prioritized for theory and experiments.

Introduction: ML in Nuc/Astro

Machine learning has already been widely explored in both nuclear physics and astrophysics.

Nuclear Physics

- **Nuclear theory**
  DFT emulation [1]
- **Calorimetry**
  GlueX [2]
- **Particle identification**
  DeepRICH [3]
- **Accelerator control**
  Real-time tuning [4]

Nuclear Astrophysics ⟷

Astrophysics

- **Cosmology**
  Moriwaki et al. [5]
- **Galaxy evolution**
  Fraser et al. [6]
- **Cosmic-ray analysis**
  Arimura [7]
- **Stellar abundances**
  den Hartogh et al. [8]

However, ML applications in nuclear astrophysics remain limited [9]; integrating AI with experiments is even less explored.

Reference

Part I: AI for Scientific Research

Robin: a closed lab-in-loop workflow

Scientific question: Find potential treatments with drugs for dry age-related macular degeneration (dAMD).

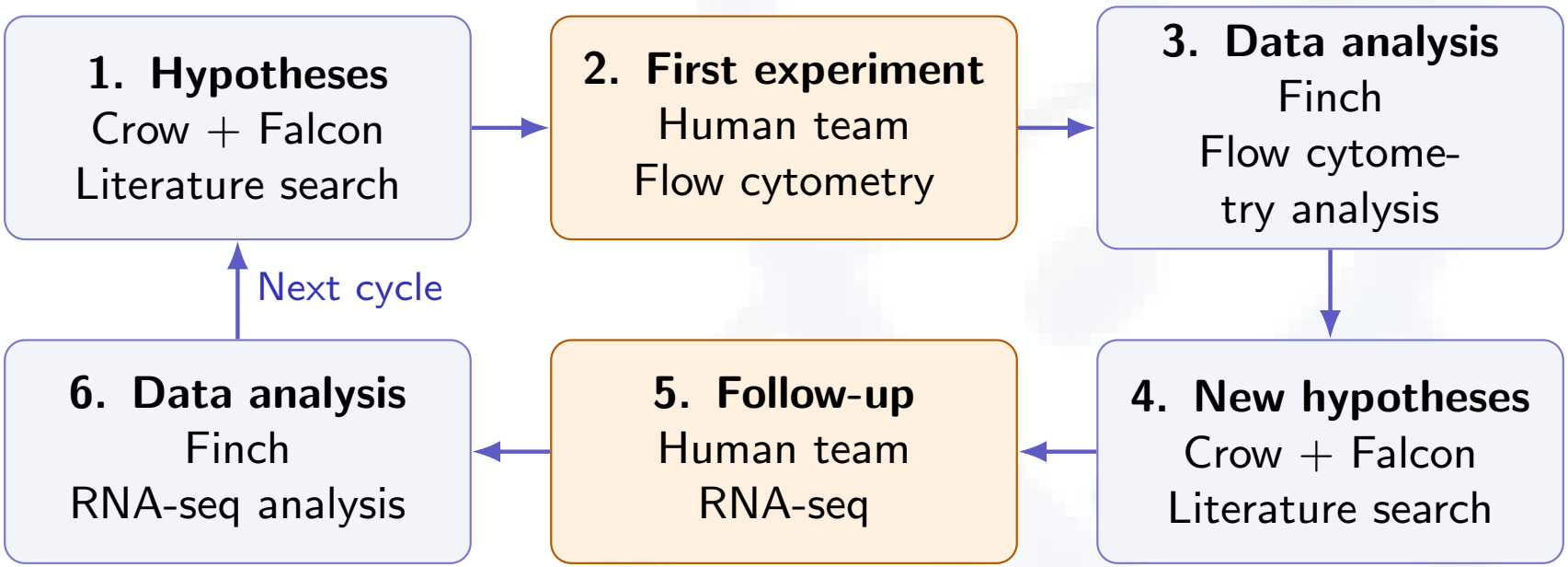


Figure 1: Robin's research loop. Steps 4–6 echo the logic of **Bayesian design**.

Findings: Ripasudil(best currently) and KL001(new mechanism) outperform; Follow-up RNA-seq implicated ABCA1 gene in the ripasudil response.

Robin, Nature (2026)

Techniques

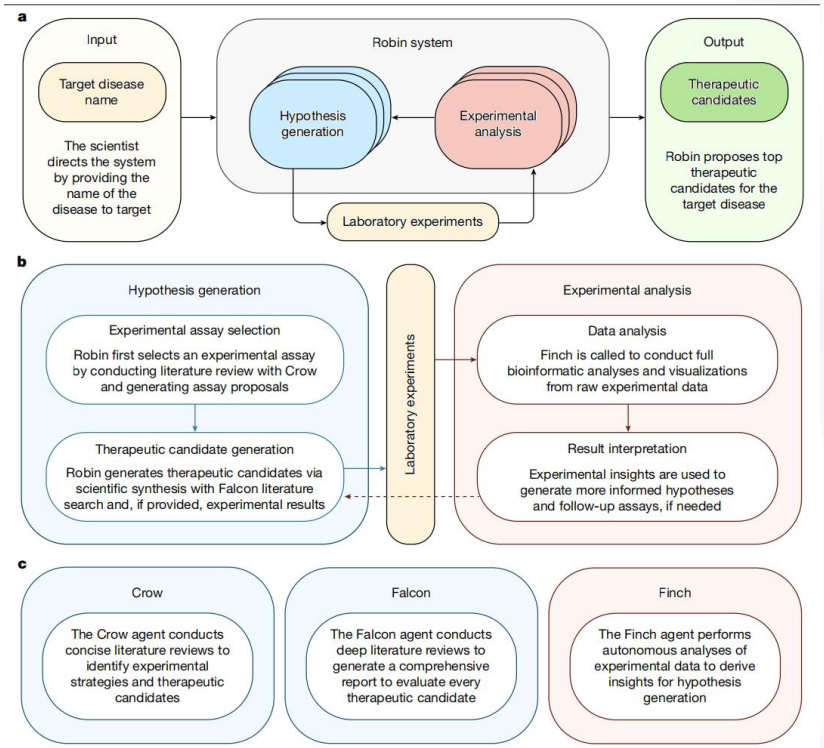


Figure 2: Robin architecture

MAS

- Crow / Falcon: PaperQA2
- Finch: Jupyter kernel

Evaluation

On position bias:

- Tournament
- Bradley–Terry–Luce (BTL)

Prompt engineering

- Iteration

Robin, Nature (2026)

Co-Scientist: focus on generating hypotheses

Core reasoning steps

1. Multi-agent self-play scientific debate
2. Tournament-based hypothesis comparison and ranking
3. Evolution with meta-review and refinement

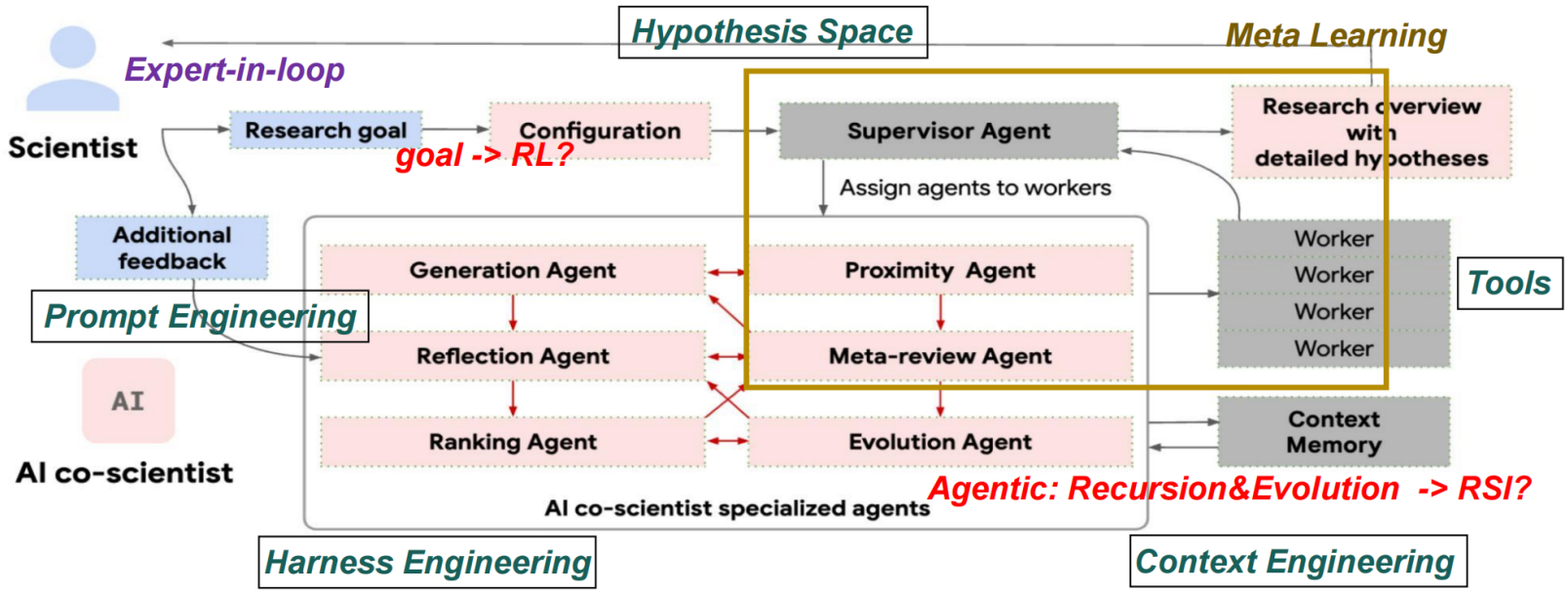


Figure 3: Co-Scientist architecture with annotated research concepts.

Co-Scientist, Nature (2026)

Co-Scientist: test-time compute scaling

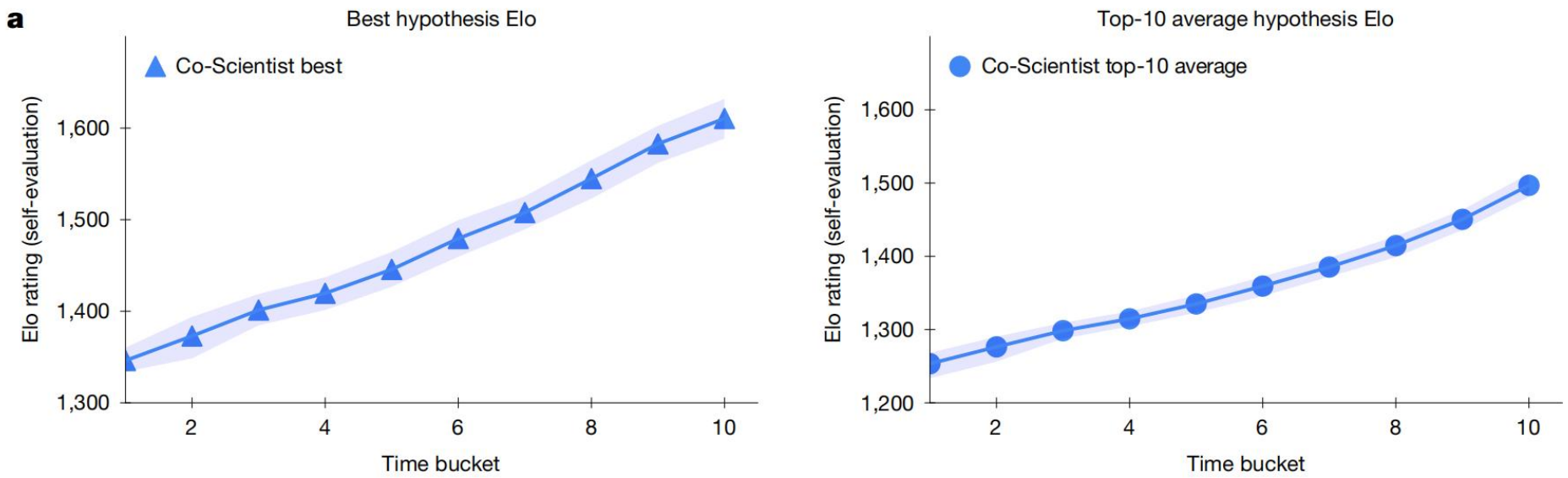


Figure 4: Internal Elo ratings of the best and top-ten hypotheses over time.

Observation: More inference-time search improves internally ranked hypotheses.

Limits

1. No saturation was observed within the tested compute range.
2. No general quantitative law in FLOPs, tokens, or cost was established.
3. Internal Elo ratings cannot replace experimental success rates.

Co-Scientist, Nature (2026)

Open Question: Emergence

The paper's perspective

The authors interpret persistent context and iterative self-improvement as an **emergent internal model of the scientific research process**.

- **Knowledge integration:** Connect accumulated knowledge, synthesize information, identify gaps, and guide further hypothesis exploration.
- **Meta-review feedback:** Distil patterns from reviews and debates into subsequent prompts, improving generation and critique without parameter updates.
- **Adaptive orchestration:** The Supervisor uses progress statistics to allocate compute across agents and guide subsequent search.

Open question

Can scientific intelligence emerge from this evolving internal model of the research process?

Co-Scientist, Nature (2026)

Co-Scientist: focus on generating hypotheses

Default Criteria

1. **Alignment** with the research goal.
2. **Plausibility:** Check reasoning and scientific assumptions.
3. **Novelty:** Go beyond existing knowledge; explain mechanisms.
4. **Testability:** Validate empirically through experiments.
5. **Safety:** Control outputs to prevent unsafe or harmful research.

Additional criteria can be configured.

Co-Scientist, Nature (2026)

Scientific Exploration

Search + Optimization

A **combinatorial search and optimization problem** in hypothesis space

Exploration $\longleftrightarrow$ Exploitation

Balance finite compute between new ideas and refinement.

Co-Scientist, Nature (2026)

New techniques to be explored...

Technique	Role and possible application
World model	**Predict observable consequences** of hypotheses to support reasoning and experiment planning.
Reinforcement learning	**Generate hypotheses** using rewards from scientific evaluation and experiments.
Bayesian design	**Guide the experimental loop:** select informative tests and update beliefs from results.
Causal discovery	**Infer causal relationships** through data and interventions.
Multi-objective search	**Select hypotheses** by balancing novelty, plausibility, testability, and cost.
Continual learning	**Self-train the model** on validated hypotheses and research trajectories.

Co-Scientist, Nature (2026)

XunZi: multi-model data fusion

Motivation: Reduce hallucinations with multimodal empirical data.

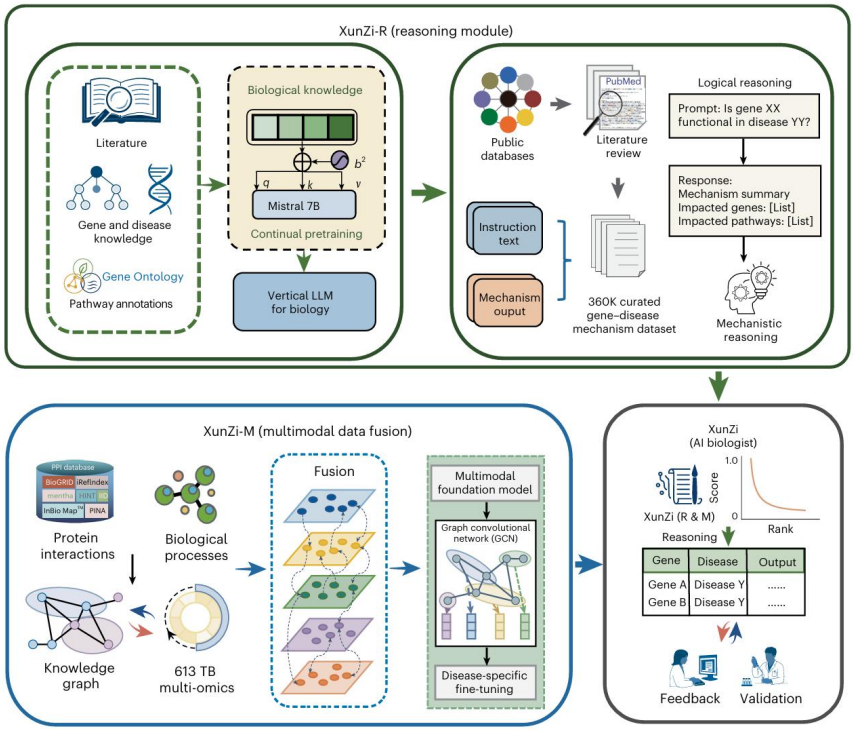


Figure 5: XunZi architecture

XunZi-R
Reasoning model
Continual pretraining
+ Supervised fine-tuning

XunZi-M
GCN fusion model

Dataset

1. Publications
2. Structured biological data
3. Mechanism CoT (manually checked)

XunZi, Nature (2026)

XunZi: multi-model data fusion

XunZi-R: Mistral-7B

- **CPT:** 24.41M PubMed abstracts + 2.05M structured knowledge records; next-token prediction.

$$\mathcal{L}_{\mathrm{CLM}} = -\sum_{t=1}^{T} \log P(x_t \mid x_{<t}; \theta)$$

- **SFT:** 330,000 gene–disease mechanism examples.

$$\mathcal{L}_{\mathrm{SFT}} = -\sum_{t=1}^{T} \log P(y_t \mid y_{<t}, x; \theta)$$

- **Output:** Confidence s for feature fusion, plus a mechanistic explanation for interpretability.

XunZi, Nature (2026)

XunZi: multi-model data fusion

XunZi-M: Two-layer GCN

- **Data:** 2.81M PPI records + 47,900 GO terms;
- **Graph:** 621,000 nodes and 6.09M edges.
- Predict disease-specific gene relevance with binary cross-entropy.

$$\mathcal{L}_{\mathrm{BCE}} = -\sum_{i \in \mathcal{Y}} [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]$$

- **Output:** Continuous scores in $[0, 1]$ are used to rank candidate targets.
- **Fusion:** Concatenate graph features with s; a linear layer produces the final relevance score.

XunZi, Nature (2026)

IdeaScientist: RL in hypothesis generation

- **Insight:** Transfer mechanisms that solved analogous challenges in other fields.
- **Three roles:** Gap finding, innovation, and report writing; each trained with reinforcement learning.
- **Corpus:** Svalbard Idea Vault, 2.77M decomposed research ideas.
- **Evaluation:** Train before a temporal cutoff; test after it.

Natural intersection of our research

Particle physics, nuclear physics, astrophysics, cosmology.

IdeaScientist, arXiv:2610.04074 (2026)

IdeaScientist: RL in hypothesis generation

The Svalbard Idea Vault

- An offline corpus of **2.77 million decomposed ideas**.
- Each tuple contains problem, challenge, gap, intuition, and proposal.
- Papers from 1986 to June 18, 2026; reference checks and topic-relevant method filtering.
- **Inputs:** Problem definition and challenge.
- **Three targets:** Methodological gap, research intuition, and complete proposal.

IdeaScientist, arXiv:2610.04074 (2026)

IdeaScientist: RL in hypothesis generation

Reinforcement Learning

- A separate LoRA adapter for each role; asynchronous GRPO with role-specific rubric rewards.

$$r_{\text{rubric}} = g\left(\lambda_{\text{free}} r_{\text{free}} + \lambda_{\text{ref}} r_{\text{ref}} + \lambda_{\text{cit}} r_{\text{cit}}\right)$$

$$\boxed{r = 0.85 r_{\text{rubric}} + 0.15 r_{\text{proc}}}$$

- **Reference-free:** LLM judges the output itself for methodological specificity and significance, feasibility, evidence, and clarity.
- **Reference-grounded:** Compare key fields (e.g., limitations, method, evaluation) with the reference, allowing valid alternatives.
- **Citation F1:** Deterministic F1 of the generated and reference citation sets, balancing precision and recall.

IdeaScientist, arXiv:2610.04074 (2026)

Other papers: complementary capabilities

System / publication date	Contribution and paper link
The AI Scientist Sakana AI et al. 25 March 2026	Agentic code-tree search within an end-to-end research workflow. Nature: 10.1038/s41586-026-10265-5
Paper2Agent Stanford 16 September 2026	Converts papers, code, and data into tested MCP tools and interactive agents. Nature: 10.1038/s41586-026-11044-y
ERA Google 19 May 2026	LLM code revision and tree search optimize empirical scientific software. Nature: 10.1038/s41586-026-10658-6
AI X-ray Scientist SLAC / Stanford et al. 1 July 2026	Tool-using agent for crystal alignment: simulated beamline to real instrument, with multimodal observation feedback. NMI: 10.1038/s42256-026-01261-5

Reference

Part II: ML in Nuclear Astrophysics

Exp Optimization: Online Bayesian Optimization

Scientific question

Can beam-based Bayesian optimization align SECAR and improve mass-plane optics with fewer measurements than manual tuning?

ML problem: Sample-efficient optimization of measured beam metrics.

Mathematical Abstraction

Alignment

$$\mathbf{u}_s^* = \arg\min_{\mathbf{u}_s} d_{\mathrm{FP1}}(\mathbf{u}_s)$$

Mass-plane optics

$$f_{\mathrm{FP2}}(\mathbf{u}_m) = \left(\frac{x + w_x - x_0}{20}\right)^4 + \left(\frac{w_y}{70}\right)^4$$

$\mathbf{u}_s, \mathbf{u}_m$: steerer and multipole settings; d_{FP1}: spot-center spread.

x_0: beam axis; w_x, w_y: spot widths.

Miskovich et al., PRAB 25, 044601 (2022)

Online Bayesian Optimization: Method and Results

Technical Stack

- **Diagnostics:** FP1 spot centers; FP2 position and widths.
- **Surrogate:** A Gaussian process, updated after each measurement.
- **Next setting:** Minimize $\mathrm{LCB}(\mathbf{u}) = \mu(\mathbf{u}) - \xi\sigma(\mathbf{u})$.
- **Control:** GPy / GPyOpt + PyEpics; steerers, then multipoles.

Results

- Angular deviation < 1 mrad.
- At least $3\times$ faster than manual tuning.
- 32% smaller beam-spot width for ^{133}Cs.

Miskovich et al., PRAB 25, 044601 (2022)

Exp Optimization: SECAR (p, n) Optics

Scientific question

Which SECAR settings separate nearly isobaric (p, n) recoils from the beam while retaining enough recoils for a cross-section measurement?

ML problem: Simulation-guided multi-objective optics optimization.

Mathematical Abstraction

$$\mathbf{u} \in \mathbb{R}^{19} \xrightarrow{\text{COSY-INFINITY}} (\Delta_{\mathrm{FP2}}, \Delta_{\mathrm{FP3}}, w_1, \ldots, w_K)$$

$$\text{maximize} \quad (\Delta_{\mathrm{FP2}}, \Delta_{\mathrm{FP3}})$$

$$\text{minimize} \quad (w_1, \ldots, w_K)$$

Δ: recoil–beam separation at slits; w: beam width along the separator.

Tsintari et al., PRR 7, 013074 (2025)

SECAR (p, n) Optics: Method and Results

Technical Stack

- **Search:** 19 field settings; recoil separation and beam widths.
- **Simulation:** COSY-INFINITY beam and recoil transport.
- **Optimization:** MOEA/D, Tchebycheff decomposition, differential evolution, and polynomial mutation.
- **Validation:** Beam tests of Pareto-selected optics; GP transmission interpolation.

Results

New ion optics enabled an astrophysical (p, n) measurement:

$$^{58}\mathrm{Fe}(p, n)^{58}\mathrm{Co}, \qquad \boxed{\sigma = 20.3 \pm 6.3 \text{ mb}}.$$

Tsintari et al., PRR 7, 013074 (2025)

Deepfield

An AI-driven Nuclear Astrophysics Laboratory

Name: Why Deepfield?

Deep $\rightarrow$ nuclear **Field** $\rightarrow$ astrophysics

Deep / Field also evokes Jinping's deep underground laboratory.

The Hubble Deep Field experiment (HDF)

In 1995, Robert Williams, then director of the Space Telescope Science Institute, proposed a different idea:

What would happen if we pointed the most advanced telescope at a seemingly empty patch of sky and observed it for long enough?

(Ursa Major) **10 days** $\longrightarrow$ **$\sim$3,000** galaxies

Significance: From searching for ever-known targets to explore unknown regions——
So how do strategies and resource allocation shape discovery?

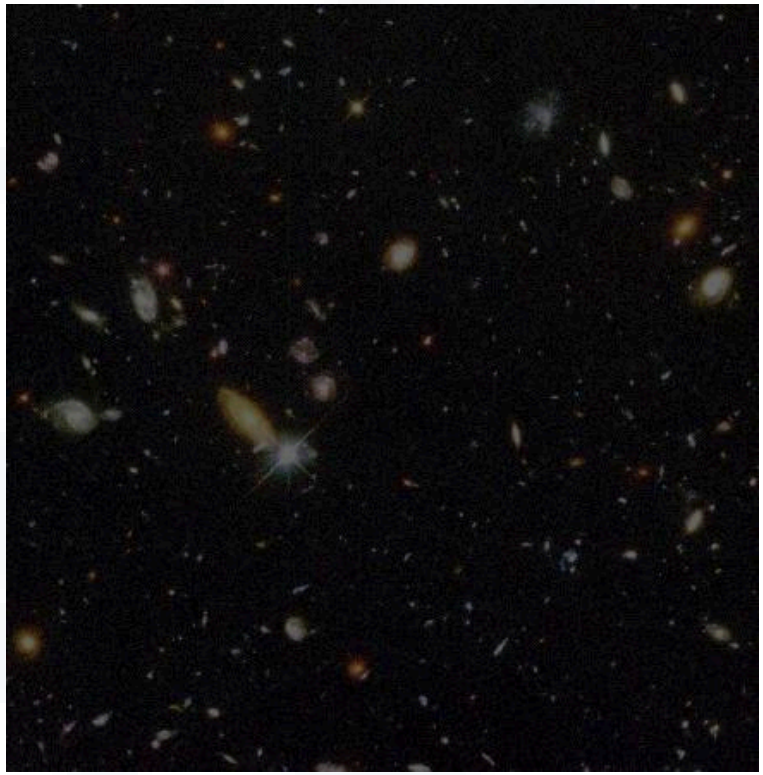


Figure 6: Central portion of the HDF

NASA, Robert Williams, and HDF Team (STScI).

NASA Science: Hubble's Deep Fields

Deepfield Research Proposal

An AI-Driven, Human-in-the-Loop Laboratory

Setting: JUNA / Jinping deep underground nuclear astrophysics experiments.

Research Vision

1. **Autonomous research:** Investigate literature, generate hypotheses, reuse and develop code, and design and analyze experiments.
2. **Human-in-the-loop:** Scientists conduct experiments; the full research process remains understandable, inspectable, and explainable.
3. **Scientific loop:** Connect literature, hypotheses, simulations, nuclear experiments, and astronomical observations in one iterative loop.
4. **Recursive improvement:** Learn better research strategies and recursively improve agent capability and efficiency by meta-learning and iteration.
5. **Long-term aim:** Progress from semi-autonomous to autonomous research, and explore the evolution and emergence of scientific intelligence.

Acceptance Criteria

For one scientist-confirmed reaction question

1. Improve on our current findings with additional promising energy regions or more informative competing hypotheses.
2. Complete a physical cross-section measurement and reproducible analysis.
3. Have the system proactively propose discriminating experiments, astronomical observations, archival analyses, or simulations.
4. Complete the follow-up with the responsible teams and independently verify its results.
5. Compare astrophysical conclusions before and after measurement.

Acceptance requires: Physical consistency checks (orders of magnitude, dimensions, conservation laws, etc.); real experimental validation; and additional, independently verified scientific findings.

Example Scientific Loop

Illustrative workflow; not an established discovery

1. **Literature:** Find a reaction, key energy regions, and conflicting evidence.
2. **Hypothesis:** Propose competing resonance mechanisms.
3. **Code:** Reproduce calculations and predict cross sections and astrophysical consequences.
4. **Experiment:** Measure at JUNA; extract cross sections, rates, and uncertainty.
5. **Follow-up:** Code proposes a discriminating test; run targeted simulations or analyze new / archival observations after coordinated execution.
6. **Interpretation:** Check results independently; return new constraints and unresolved questions to the next cycle.

MAS Roles

Core roles

- Literature
- Hypothesis
- Code
- Experiment
- Verification
- Supervisor

Hypothesis refinement

- Reflection
- Ranking
- Proximity
- Evolution
- Review synthesis

Towards an AI co-scientist, arXiv:2502.18864

Data Engineering

- **Research objects:** Link papers, nuclear data, code, measurements, simulations, and observations through persistent identifiers.
- **Grounded extraction:** Trace formulas, tables, plots, and claims to source locations; check extracted values against the originals.
- **Scientific metadata:** Encode reactions, units, energy frames, validity, and uncertainty/covariance; distinguish measurements from predictions.
- **Agent access:** Combine metadata filters with text and image retrieval; return focused evidence and links to complete artifacts.
- **Learning datasets:** Curate reviewed research trajectories and failures; separate training, optimization, and frozen tests by source, time, and reaction.

FAIR, Scientific Data (2016); Paper2Agent, Nature (2026)

Harness Engineering

- **Adaptive orchestration:** Delegate bounded tasks; revise plans and parallelism according to dependencies, evidence, and computational cost.
- **Durable execution:** Track job IDs, checkpoints, dependencies, and artifact versions; resume long jobs without duplicating actions.
- **Shared workspace:** Keep tasks, evidence, branches, and run histories inspectable; give specialists focused contexts and explicit handoffs.
- **Feedback control:** Use tool results, physical checks, and human experimental feedback to revise plans, recover failures, or stop unproductive branches.
- **Evaluate and evolve:** Test orchestration changes on held-out tasks for scientific value, reliability, human effort, and cost.

Co-Scientist, Nature (2026); Agent harnesses, Anthropic (2025)

Prompt Engineering

- **Task contracts:** Specify scientific goals, assumptions, constraints, available tools, and measurable success criteria.
- **Research outputs:** Request predictions, evidence links, uncertainty, and falsification tests in schemas matched to each role.
- **Scientific exploration:** Generate competing mechanisms, seek counterevidence, and choose informative tests; explain decisions with evidence.
- **Reusable guidance:** Supply concise examples and tested procedures; retrieve specialized instructions when relevant.
- **Reflective optimization:** Use traces, failures, and expert feedback to evolve prompts with methods such as GEPA; test on held-out reactions.

GEPA (2025); ERA, Nature (2026)

Context Engineering

- **On-demand context:** Store complete artifacts externally; load evidence, code, and tool results as the current task requires.
- **Memory layers:** Separate working state, research episodes, and reusable strategies; attach source, version, and validation status.
- **Scientific fidelity:** Preserve values, units, uncertainty/covariance, counterevidence, and unresolved disagreements during compression and handoffs.
- **Incremental learning:** Reflect on outcomes and curate validated lessons; update or retire individual entries rather than rewriting all memory.
- **Transfer and evaluate:** Share artifact references and open questions across agents; measure memory's benefit on held-out research tasks.

ACE (2025); Context engineering, Anthropic (2025)

Tool Use

Invocation loop

Goal → discover → execute → verify → update.

- **Discover on demand:** Find relevant literature, data, simulation, and instrument tools; load only the needed schemas and examples.
- **Validated reuse:** Reproduce paper workflows; package reusable tools with fixed environments, reference datasets, and regression tests.
- **Scientific contracts:** Specify inputs/outputs, units, energy frames, valid regimes, uncertainty, versions, and expected costs.
- **Execution feedback:** Return structured results and diagnostics; check numerical and physical validity before interpreting outputs.
- **Tool evolution:** Use failure traces to improve descriptions, wrappers, and tests; evaluate changes on independent scientific tasks.

Paper2Agent, Nature (2026); Advanced tool use, Anthropic (2025)

Multimodal Fusion and Model Training

- **Grounded fusion:** Connect text, equations, spectra, nuclear measurements, simulations, and observations through shared physical quantities.
- **Quantitative reasoning:** Couple language reasoning with physical solvers and statistical inference; propagate measurement and model uncertainty.
- **Workflow learning (SFT):** Learn scientific workflows from expert-reviewed, reproducible trajectories; use LoRA for efficient adaptation.
- **Hypothesis learning (RL):** Improve hypothesis generation using feedback on novelty, physical plausibility, testability, and supporting evidence.
- **Continual learning:** Self-train on newly validated trajectories and results; test transfer, retention, and gains beyond retrieval or prompt updates.

XunZi, Nature Biomedical Engineering (2026); IdeaScientist, arXiv:2610.04074 (2026)

Validation

- **Executable outcomes:** Reproduce reference results; check convergence, regression cases, and actual outputs against task requirements.
- **Physical checks:** Verify units, dimensions, orders of magnitude, conservation laws, valid regimes, and known limiting cases.
- **Uncertainty checks:** Propagate errors and correlations; assess interval coverage, model sensitivity, and robustness to new data.
- **Scientific evidence:** Test competing predictions with real experiments and astronomical observations; independently verify new findings.
- **Feedback and revalidation:** Diagnose failures, revise the responsible components, and retest on held-out tasks for scientific value, reliability, and cost.

Co-Scientist, Nature (2026); Agent evals, Anthropic (2026)

Six Research Directions

Direction	Research question
I. Knowledge and models	How can new evidence improve physical models and research decisions?
II. Hypothesis generation	How can agents propose and refine new, testable physical explanations?
III. Analysis and simulation	How can reproducible calculations reveal what to investigate next?
IV. Experimental design	Which feasible measurement best resolves the scientific uncertainty?
V. Scientific experimentation	What do real measurements and targeted follow-ups establish?
VI. Benchmarking	What scientific gains does each method add to the full research loop?

Near term: one complete JUNA research loop. Longer term: transfer across reactions and environments, with evolving research strategies.

I. Continual Knowledge-Guided Exploration

- **Align changing evidence:** Link literature, nuclear data, measurements, and observations; reconcile units, conditions, and conflicting results.
- **Discover physical models:** Search physically constrained equations or programs; revise mechanisms when existing models fail quantitative tests.
- **Learn from experience:** Connect configurations, anomalies, successful procedures, and failures to their conditions and scientific outcomes.
- **Sustain research:** Preserve hypothesis branches, task dependencies, and pending experiments; resume work as new evidence arrives.
- **Test decision value:** Compare fixed and evolving knowledge on reaction priorities, model predictions, and proposed follow-up measurements.

Output: traceable evidence, executable predictive models, and reusable research experience.

II. Hierarchical Scientific Hypothesis Generation

- **Hierarchical doubt:** Question assumptions, approximations, validity, and anomalies across measurements, nuclei, networks, astrophysics, and fundamental physics.
- **Cross-domain reasoning:** Transfer theories, mathematical structures, and mechanisms through analogy and abduction.
- **Simulation-guided discovery:** Use Code scans, sensitivity, and uncertainty to reveal dominant factors, model mismatch, and new questions.
- **Mechanisms:** Derive competing causal explanations, observable predictions, and their uncertainty.
- **Open-world generation:** Explore unlisted variables, mechanisms, and questions; request new or revised models in Direction I.
- **Robustness and falsifiability:** Refine using both experiment types; test parameter, model, and uncertainty assumptions and state falsification conditions.
- **Quantitative selection:** Combine scientific value, novelty, information gain, sensitivity, uncertainty, and feasibility; retain complementary hypotheses.

Output: Versioned hypothesis cards with evidence, predictions, falsification tests, and revisions.

III. Data Analysis and Scientific Simulation

From reproducible computation to new questions

Discover → reproduce → reuse / extend → validate.

- **Build on existing software:** Reproduce reference cases; adapt validated tools and develop new code where capabilities are missing.
- **Infer nuclear quantities:** Model calibration, background, and efficiency; extract cross sections and rates with correlated uncertainties.
- **Connect to astrophysics:** Propagate rates through reaction networks and stellar models; quantify changes in abundances and observables.
- **Propose follow-ups:** Identify unresolved mechanisms and suggest discriminating observations, archival analyses, or controlled simulations.
- **Deliver reproducible results:** Preserve data, code, environments, physical checks, and the calculations behind each conclusion.

Paper2Agent, Nature (2026); ERA, Nature (2026)

IV. Experimental Design and Optimization

- **Define the scientific utility:** Target model discrimination, parameter precision, or reduced uncertainty in astrophysical predictions.
- **Choose informative measurements:** Use Bayesian experimental design to select energies, observables, and exposure time under realistic constraints.
- **Optimize instrument performance:** Use simulation and Bayesian optimization for beam settings, efficiency, background, and resource trade-offs.
- **Adapt from feedback:** Update physical inference with new data; revise the next measurement and stop when expected benefit no longer justifies cost.
- **Prepare feasible follow-ups:** Test plans in simulation or replay; coordinate calibration, beam time, and observation or archival access.

Output: executable plans with predicted scientific benefit, uncertainty, resource needs, and criteria for adjustment.

Bayesian experimental design; SECAR online optimization (2022); SECAR measurements (2025)

V. Scientific Experimentation

- **Nuclear measurements:** Scientists conduct beam experiments with agent assistance; acquire calibrated data, background controls, and independent checks of cross sections.
- **Astronomical observations:** Test nuclear-constrained predictions against new observations or independent archives; account for astrophysical and observational uncertainties.
- **Numerical experiments:** Vary rates and mechanisms under controlled assumptions; test convergence and propagate uncertainty to astrophysical predictions.
- **New experimental approaches (challenge):** Explore novel measurement methods and configurations to access previously inaccessible reactions or energy regions; test feasibility and scientific benefit.

Output: real measurement results, targeted follow-up results, and independently checked physical interpretations.

VI. Scientific Discovery Benchmarks

Task	Observable evaluation criteria
Literature	Evidence coverage and citation support; numerical, unit, and figure-reading accuracy; conflict detection.
Hypotheses	Independently checked novelty and physical plausibility; quantitative, discriminating predictions; revision after new evidence.
Analysis and simulation	Reference error, convergence, physical consistency, uncertainty coverage, and reproducibility.
Experiments	Feasible plans, measurement precision per beam hour, reliable execution, and realized scientific benefit.

Full-loop outcome: Additional verified findings, stronger constraints, or better discrimination between physical explanations.

Set task-specific tolerances with experts before evaluation; use raw data, reference calculations, and external evidence.

Independent Evaluation and Ablation

- **Independent tests:** Hold out by publication time, reaction system, or experimental batch; check overlap.
- **Baselines:** Human workflows, baseline Agents, full MAS; match resource budgets.
- **Ablations:** Tools, memory, fusion, and candidate roles; include coordination cost.
- **Key comparison:** Measurement and analysis alone versus the full Loop with proactive follow-ups.
- **Report:** Physics checks, calibrated uncertainty, evidence, human effort, and costs.

Separate replay, simulation, and prospective physical results. Internal rankings and analysis consensus are not independent evidence.

Seven-Month Research Roadmap

Month	Focus and milestone
1	**Literature:** Evidence, confirmed pilot, and beam-time coordination.
2	**Code:** Reproduce at least one reference within expert tolerances; reusable tools and a discriminating follow-up proposal.
3	**Hypothesis:** At least two distinguishable explanations; quantitative predictions and falsification criteria.
4	**Experiment:** Approved plan; optimization and instrument-feedback demo.
5	**Beam:** Real cross sections and rates; statistical and correlated systematic uncertainties.
6	**Follow-up:** Execute the second experiment; compare astrophysical conclusions before and after measurement.
7	**Integration:** Frozen independent benchmarks, matched-budget baselines and ablations, research report.

Develop Code tools before Hypothesis; the scientific Loop stays iterative. Shared engineering throughout; monthly benchmarks and frozen versions. Beam scheduling from Month 1. Replay tests capability; delayed real measurements leave the Loop pending.

Seven-Month Research Budget

CNY 100,000; single-developer project

Item	CNY
Large language model APIs	40,000
CPU and GPU computing	25,000
Expert review, annotation, verification	20,000
Retrieval, OCR, embeddings	5,000
Contingency	10,000
Total	**100,000**

Expert services are combined; CPU and GPU resources are used on demand.

Excludes developer compensation, beam experiments, experiment-related travel, and storage.

Thank You

Questions and Discussion

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